Spatial ML - class 5 - spatial Random Forest

Total points 100/100

Email *

o.marvan@student.uw.edu.pl

Student index: *

477001

Name: *

Ondřej

Surname: *

Marvan

What is spatial autocorrelation? *

10/10

- ( ) Data follow a normal distribution
- ( ) Data are randomly distributed in space
- (•) Nearby observations tend to have similar values
- ( ) Variables are independent

What is the main purpose of including spatial variables in a model? * 10/10

- [x] To improve prediction and remove spatial structure from residuals
- [ ] To reduce computation time
- [ ] To standardize variables
- [ ] To eliminate multicollinearity

In Random Forest, what does bootstrap sampling mean? * 10/10

- [ ] Selecting only important variables
- [ ] Using all observations in every tree
- [ ] Sampling without replacement
- [x] Sampling observations with replacement for each tree

What is the final prediction in a Random Forest regression model? * 10/10

- (●) The average of predictions from all trees
- ( ) The prediction from the best tree
- ( ) The minimum prediction with best AIC
- ( ) The maximum prediction

What does %IncMSE measure in variable importance? * 10/10

- ( ) Correlation between variables
- ( ) Number of splits in trees
- (●) Increase in prediction error after permuting a variable
- ( ) Model accuracy

What does minimal depth indicate? *

10/10

- ○ Number of trees in the forest
- ○ Depth of the deepest node
- ◉ How early a variable appears in a tree
- ○ Number of observations

What is the role of mtry in Random Forest? *

10/10

- ○ Number of terminal nodes
- ○ Number of trees
- ◉ Number of variables randomly selected at each split
- ○ Number of observations

What does the following R code do? * 10/10

rf.global <- randomForest(x, y, importance=TRUE)

- ○ Plots a decision tree
- ○ Creates a linear regression model
- ◉ Builds a Random Forest model and computes variable importance
- ○ Standardizes the dataset

What is the purpose of the following code? * 10/10

partialPlot(rf.model, x, locPdens)

- ○ Shows correlation matrix
- ○ Displays distribution of variables
- ◉ Shows marginal effect of locPdens on prediction
- ○ Tunes hyperparameters

What is the purpose of the following R code? *

10/10

my.tune <- tune(eq, data=firms.sel)

- ◯ Removes spatial autocorrelation
- ◯ Calculates variable importance
- ◉ Performs hyperparameter tuning for the Random Forest model
- ◯ Fits a single decision tree